

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(d)		$\ln 119 = 4.78$ or $\ln 87 = 4.47$ (1) $\ln 103 = 4.63$ or the other extreme (1) $4.78 - 4.63$ or $4.66 - 4.47 = [\text{approx. } 0.15]$ or $\frac{0.31}{2}$ (1) No ecf allowed (since 103 and 16 are given) Award 1 mark for $\frac{16}{103} = 0.15, 0.155, 0.16$ Award 2 marks for statement absolute uncertainty in log = fractional uncertainty in count		3		3	3	3
	(e)	(i)	Taking logs: $\ln(N) = -n \ln(r) + \ln(k)$ (1) (accept $\ln I$ in lieu of $\ln N$) Plotting $\ln(N)$ on the y-axis against $\ln(r)$ on the x-axis or comparison with $y = mx + c$ (1) Gradient = $-n$ (1)			3	3	3	3
		(ii)	Minimum gradient = $\frac{3.94-7.00}{0.50-(-1.30)} = [-]1.69 \pm 0.03$ (1) Maximum gradient = $\frac{3.40-7.00}{0.50-(-1.17)} = [-]2.16 \pm 0.03$ (1) Mean gradient = $[-]1.93 \pm 0.03$ (1) ecf Uncertainty = $\frac{1}{2} (-1.69 - (-2.16)) = 0.20 - 0.28$ (1) ecf accept 0.2 No sig fig penalty		4		4	4	4
			Question 5 total	0	14	7	21	17	21